

Full Length Article

Dopamine-assisted sustainable antimicrobial peptide coating with antifouling and anticorrosion properties

Pan Cao^a, De Liu^a, Yanbing Zhang^b, Fei Xiao^a, Chengqing Yuan^c, Feng Liang^a, Xiaodan Liu^c, Chao Zhang^{a,*}^a School of Mechanical Engineering, Yangzhou University, Yangzhou 225127, China^b College of Animal Science and Technology, Yangzhou University, Yangzhou 225009, China^c Reliability Engineering Institute, National Engineering Research Center for Water Transport Safety, Wuhan University of Technology, Wuhan, 430063, China

ARTICLE INFO

Keywords:

Dopamine treated antimicrobial peptide
Surface modification
Antifouling surfaces
Antibacterial performance
Corrosion resistance

ABSTRACT

Surface biofouling will seriously affect the performance of solid–liquid contact surfaces. Antimicrobial peptides (AMPs) are considered as potential agents for surface antifouling due to its broad-spectrum antimicrobial capacity. In this study, a derived AMP was synthesized by the reaction between a marine originated AMP and monomolecular dopamine, which was then used to modify 304 stainless steel. Mass spectrometry analysis confirmed the synthesis reaction of DP, and the co-culture essays with E11 and 293 T cells proved DP's excellent compatibility. Fourier transform infrared spectroscopy and X-ray photoelectron spectroscopy analysis confirmed that DP was grafted on surface successfully, meanwhile, atomic force microscope, contact angle and 3D optical profilometer results demonstrated that surface morphology and wettability changed remarkably after modification. The results of *Vibrio natriegens* and *Phaeodactylum tricornutum* adhesion assays illustrated that DP treated surface possessed great antifouling capacity, and the antibacterial and anti-algal properties of the DP modified surface can reach 97.78% and 90.01%, respectively. Electrochemical results indicated that DP modified surface exhibited superior anticorrosion. The stability test results showed that DP modified surface possessed long-lasting properties in antifouling and anticorrosion. These findings may provide a useful tethering process for producing excellent antifouling surfaces in marine equipment, medical devices and other relevant fields.

1. Introduction

Biofouling is an undesirable attachment phenomenon of fouling organisms, which widely exists on the surface of ship hull, ship structure, port facilities, medical and food processing equipment [1–3]. Marine bioorganisms accumulation will increase the ship's own weight, reduce the speed loss and fuel efficiency of the ship, and even cause serious corrosion, resulting in increased costs and CO₂ emissions [4]. Marine biofouling have prompted scientists and engineers to develop antifouling coatings, which are mainly relies on the continuous release of antifoulants or the release of toxic metal ions to inhibit/kill bioorganisms near the surface of the hull [5,6]. However, the toxic substances released from antifouling coatings will accumulates for a long time, causing serious harm to the marine ecology [7]. It's believed that adsorption of organic molecules and biofilm formation are the initial stage of biofouling, biofouling could be refrained by reducing the formation of conditioning film and biofilm [8,9].

AMPs, as a kind of organic matter derived from nature, have non-toxic, green, and broad-spectral antibacterial properties with great potential in the field of surface antifouling [10]. In recent years, more studies have been performed to surface modification of antimicrobial peptides for tackling surface antifouling [2,11–13]. Wong *et al.* generated a novel antifouling surface with low surface energy via modification by peptide [14]. Davis *et al.* believed that the modification was a chemical reaction between the disulfide bonds of the peptide and the iron ions of stainless steel by sharing electrons [15]. Cao *et al.* discovered that the wettability and antibacterial properties of the stainless steel surface modified with AMPs had undergone significant changes, and the surface had an obviously inhibitory effect on *Escherichia coli* and *Staphylococcus aureus* [16]. Mondal *et al.* reported antibacterial surface strategy via a combination of a gasotransmitter, NO, and nisin (an AMP) on SR polymer surface, achieving long-term antimicrobial efficacy and biocompatibility properties [17]. Xiao *et al.* combined the sum frequency generation (SFG) vibrational spectroscopy and the ground

* Corresponding author.

E-mail address: zhangc@yzu.edu.cn (C. Zhang).<https://doi.org/10.1016/j.apsusc.2022.153019>

Received 13 January 2022; Received in revised form 2 March 2022; Accepted 3 March 2022

Available online 9 March 2022

0169-4332/© 2022 Elsevier B.V. All rights reserved.

microscope to observe the structures and activity of surface-immobilized AMPs, the surface-immobilized peptides changed orientation and killed bacteria [18,19].

Many methods of AMPs modification on solid surface have been studied, including monomeric self-assembly (SAMs) [20], chemical coupling [21], layer-by-layer assembly (LBL) [22], polymer brushing [23], etc. Although the above methods have achieved well modification purposes, they lack the broad spectrum property of modifying the substrate. Dopamine is oxidized and aggregated into polydopamine (PDA) in an alkaline solution under the action of oxygen, which allows PDA to activate almost all types of surfaces and has great potential for surface modification [24–26]. PDA has become a commonly used precursor for surface-bound organics, including antimicrobial peptides, which because many functional groups (catechol, amine, etc.) were produced during the oxidative polymerization of dopamine to PDA [12,27,28]. Both polymyxin B and E are AMPs, which were immobilized onto silicone rubber surface with PDA, and the peptide-modified surfaces showed great anti-adhesive and antibacterial activities for *Pseudomonas aeruginosa* under at very low AMPs concentrations [29]. Yu et al. developed a simple, one-step, substrate-independent antifouling surface by conjugating AMPs and PDA, the coating has shown to be highly effective in preventing biofilm formation [10]. In addition to AMPs, nanoparticles [30], PEI [31], GPEG [32] and OEG [33] were also used to be grafted on the substrate for antifouling applications via the PDA platform. However, there is still unclear that the exact mechanism of dopamine polymerization and the interaction between AMPs and PDA platform is not enough for applying peptide-immobilized surface in the actual marine environment [34,35]. The new method of grafting AMPs with dopamine/PDA needs to be further studied.

Stainless steel (SS) has been used in many fields due to its excellent corrosion resistance and mechanical properties. In this study, a marine antimicrobial peptide extracted from suckers of *Octopus vulgaris* was applied to modify SS surface via dopamine for antifouling applications. The antimicrobial peptide was firstly reacted with dopamine monomer under weak acidic conditions to synthesize a new type of derived antimicrobial peptide. The derived peptide was then immobilized onto 304 SS surface due to the functional groups of dopamine residues. The surface properties were characterized and the antifouling performance were evaluated by marine organisms. The novel surface modification method can provide a new strategy for surface antifouling of hull and other marine equipment without polluting the surrounding environment.

2. Materials and methods

2.1. Materials and reagents

Dopamine hydrochloride (98%), N-ethyl-N-(3-dimethylamino-propyl) Carbodiimide hydrochloride (EDC, greater than 99%), 2-(N-morpholino) ethanesulfonic acid buffer (MES, pH = 6.0), Propidium iodide (PI), Glutaraldehyde (2.5%), and Absolute ethanol were purchased from Aladdin. The peptide (QAGSNKGASQKGMS, 1350.47) was purchased from DG peptides Co., Ltd (purity greater than 95%), which has good antibacterial capacity [36]. All chemicals were used without further purification.

2.2. Preparation and compatibility of derived marine AMP

Dopamine monomer modification method has been discussed in our previous studies. 20 mg peptide and 100 mg dopamine powder were dissolved in MES buffer (25 mL, 0.1 M, pH = 6.0) and stirred well. The EDC (200 mg) was then added into MES buffer, and the mixed solution was placed on a shaker (30 rpm) for 8 h under room temperature. The reacted solution was poured the dialysis membrane (MWCO = 1.0 kDa) in a weakly acidic solution environment (1 mL 35 % HCl per 1 L distilled water) to remove dopamine, EDC and MES from the mixed solution. The

acidic solution should be replaced every 3 h to maintain the exchange potential of the substances on both sides of the dialysis membrane. The derived AMP (DP) solution was obtained after 24 h dialysis, and the DP powder was generated after freeze-dried in a vacuum freeze dryer.

Compatibility of DP to marine creatures and human beings was evaluated by Striated Channa cells (E11) of Zebrafish and human renal epithelial cells (293 T). The toxicity of DP to fish and human cells was assessed by the CCK-8 kit via 0, 2, 4, 8, 12, and 24 h co-culture of cells and DP. The cell solution (2000 cells/mL) was added into 96-well cell culture plate, and incubated in an incubator (37 °C) for 24 h with 5% CO₂ volume fraction. After 2 h, CCK-8 solution was added to wells (10 µL/well) and the absorbance at 450 nm was measured by an enzyme marker. The morphology of the cells after co-culture with DP was observed and collected by a fluorescent inverted microscope.

2.3. Preparation of SS-DP

304 SS samples (10 mm × 10 mm × 1 mm) were polished by 200, 400, 800, 1200, and 2000# sandpaper in succession to obtain smooth surfaces. The samples were immersed into absolute ethanol for 15 min, and then the samples were cleaned in acetone by sonication. The blank 304 SS samples can be used after rinsing by distilled water and drying in a vacuum dryer. 304 SS were immersed into Tris-HCl solution containing 0.1 mg/mL DP for 12 h at room temperature, then the derived antimicrobial peptide treated surfaces (SS-DP) were prepared. SS-DP can be used for surface characterization and antifouling tests after cleaning by distilled water and drying under nitrogen chamber. The synthesis was analyzed by simulation of the reaction between peptide and dopamine using Comparative Molecular Field Analysis (CoMFA).

2.4. Characterization of SS-DP

The mass spectrometer of the DP was analyzed by LCMS-2020 mass spectrometer (Shimadzu, Kyoto, Japan). Fourier transform infrared spectroscopy (FTIR, NICOLET Is50) was applied to collect the information of SS and SS-DP surface in the scanning range from 4000 to 400 cm⁻¹ for 128 scans at a spectral resolution of 4 cm⁻¹. The water contact angle of the sample surfaces was measured via a contact angle system (Dataphysics OCA 15EC, Germany) with 3 µL drops. Surface elements information of the samples was obtained by X-ray photoelectron spectroscopy (XPS, ESCALAB250Xi, USA). Surface morphology of the samples was characterized via an atomic force microscope (AFM, SPM 9700, Japan). A field emission scanning electron microscope (SEM, Zeiss Ultra Plus, Germany) and a confocal laser scanning microscope (CLSM, Leica TCS SP8 STED, Germany) were utilized to evaluate the antifouling performance of the sample surfaces. Fluorescence inverted microscopy (FIM, AXIOVERT A1, Germany) was used to observe the morphology of cells on the sample surface and analyzed the toxicity of the peptide to the cells. The CHI600E electrochemical workstation with a three-electrode system was used to assess anticorrosion of surfaces of SS and SS-DP. Among the three electrodes, the saturated glycerol electrode, platinum electrode and test samples are reference electrode, auxiliary electrode and working electrode respectively. The working surface diameter of the epoxy resin encapsulated test sample is Φ10mm and the polarization curves were scanned at a rate of 5 mV/s in the range of -0.75 V to 0.5 V at room temperature using 3.5% NaCl solution as electrolyte. The binding performance of antibacterial coating on the sample surface was analyzed by thickness change. SS-DP was immersed into the culture plate containing artificial seawater (ASW), and placed the culture plate on the shaker with 30 rpm rotation to analyze the coating thickness of the sample surface.

2.5. Antifouling tests

Antibacterial test: *Vibrio natriegens* (*V. natriegens*) is a common marine bacteria widely distributed in the coastal waters of China and

was selected as a marine fouling organism to carry out surface anti-fouling experiments [12,37,38]. SS and SS-DP samples were placed into 2216E solution containing *V. natriegens* (1×10^6 CFU/mL) with 100 rpm shaking for 24 h at 37°C, the bacteria treated surfaces were then rinsed with sterile ASW solution three times to eliminate the remaining 2216E solution and bacteria that are not firmly attached. The SS and SS-DP samples were immersed into 2.5% glutaraldehyde solution for 4 h at 4 °C. Some samples were dehydrated by 25, 50, 75, 90, and 100% ethanol solutions (5 min for 25, 50, 75 and 90%, 10 min and 2 times for 100%) in turn, SEM was applied to observe the situation of attached bacteria. The other samples were stained with a sterile ASW solution containing a concentration of 50 µg/mL of PI for 30 min in the dark at room temperature. The sterilized ASW was used to remove the dye solution of the surface, and attached bacteria images were collected by CLSM. The robustness of surface antibacterial capacity were evaluated by immersing samples into *V. natriegens* solution for 7, 14 and 21 days, the bacterial solution was updated every 48 h and characterized follow the same procedure.

Antialgal test: *Phaeodactylum tricornutum* (*P. tricornutum*) is a typical marine diatoms in the East China Sea, and was selected as a fouling algae to evaluate the antialgal performance of the sample surfaces modified by AMP derivative. SS and SS-DP samples were immersed into ASW, and cultured in a constant temperature light incubator for 7, 14 and 21 days (light and dark alternately, each for 12 h) at 25°C. The algal solution was updated every 7 days. The samples were washed by sterile ASW solution to remove any unattached diatoms, and then fixed on the surfaces by immersing into 2.5% glutaraldehyde solution at 4 °C for 4 h. The attachment images of *P. tricornutum* on the sample surface can be observed and collected by a CLSM. Since diatoms contain chlorophyll, the images of *P. tricornutum* adhesion can be collected without any dyeing.

2.6. Corrosion resistance test

The corrosion resistance of the samples before and after modification was evaluated by the electrochemical workstation CHI600E and Gamry software. The sterilized SS and SS-DP were encapsulated in epoxy resin, then the circular working surface with an area of 1.0 cm² was polished, soaked in 2216E culture solution for cultivating *V. natriegens*. The samples were placed in a shaking table for culture (30 °C, 120 r/min). The samples in the corrosion experiment for testing after 1, 3 and 7 days immersion. In the electrochemical experiment, the platinum electrode was the auxiliary electrode, while the saturated glycerol electrode was the reference electrode. Samples were tested by the working electrode with the 1.0 cm² working surface. Polarization curves were generated by scanning surface in the range of −0.5 to 0.5 V at the rate of 5 mv/s at room temperature. By fitting and analyzing the polarization curve (Tafel diagram), the cathode and anode Tafel slopes of the samples before and after modification can be obtained (B_c and B_a), corrosion potential (E_{corr}), corrosion current density (i_{corr}), and instantaneous corrosion rate (CR). The inhibition efficiency (IE) of the surface-functionalized coupons was calculated using the following equation [39]:

$$IE\% = (i_{corr-SS} - i_{corr-SS-DP}) / i_{corr-SS} \quad (1).$$

The $i_{corr-SS}$ and $i_{corr-SS-DP}$ are the corrosion current densities of the original SS and modified samples, respectively, as determined by the Tafel polarization curve analysis.

2.7. Statistical analysis

In order to ensure the accuracy of the data, three quantitative measurements were taken for each group and all the difference of SS and SS-DP was determined by a standard student *t*-test ($p > 0.05$ stands for no significant, $0.01 \leq p < 0.05$ is considered statistically significant, and $p < 0.01$ is considered statistically very significant). The adhesion rate of bacteria and diatoms on the sample surface was analyzed by ImageJ software. The coverage rate was calculated by the formula: Coverage

rate = area of attached fouling organism / total view field $\times 100\%$.

3. Results and discussion

Dopamine is prone to oxidize and polymerize into PDA weakly alkaline aqueous conditions with oxygen [40,41]. A variety of intermediate products containing catechol are produced during the process, which have the ability to adhere to all surfaces, and provides the basis for the grafting of antimicrobial peptides on the surface of stainless steel [42,43]. However, the chain of PDA needs to be controlled to prevent it fall off from the substrate when it is too long, resulting in insufficient binding force of the PDA coating [24]. Considering the feature that dopamine does not self-polymerize under weak acid conditions, the dopamine monomers are first reacted with antimicrobial peptides to obtain antimicrobial peptides modified by dopamine monomers under the action of the catalyst EDC for improving grafting rate of antimicrobial peptides on the substrate [44]. DP was synthesized by conjugating the hydroxyl groups of dopamine and the carboxyl group of amino acids, and the schematic illustration of DP synthesis was shown in Fig. 1. CoMFA has been proved as a powerful 3D QSAR method in many field. The untreated AMP is regarded as the oxidized state of the reaction, while the derived peptide is in the reduced state. The reduction state energy was lower than the oxidation state energy (Fig S1D), which indicated that the occurrence of reduction exothermic reaction was relatively easier [45].

The molecular weight of DP is 1528.75 (Fig S2), compare to 1350.47 of untreated AMP's molecular weight, which is larger than the molecular weight of the resulting product if the antimicrobial peptide molecule and the dopamine monomer undergo an esterification-like reaction in an equimolar amount. Therefore, dopamine reacts with multiple sites on the untreated AMP. Dopamine bound to the AMP's molecule via the Schiff-base reaction between the amine group of dopamine (DP) and the ketone group (DP') on the AMP, and their molar ratio of the reaction is about 4:3. The total molecular energy of DP lower than DP' (Fig S1D), which also illustrated that the reaction to produce DP occurred more easily than the reaction to produce DP'. All AMP derivatives will not be separated and marked as DP in the following.

The compatibility of derived peptides was evaluated by co-culture with E11 and 293 T cells. The concentration of cells have liner relationship with the absorbance value at 450 nm, which make it possible to quantitatively analyze the amount of cells by measuring the absorbance values at 450 nm using CCK-8 kit. Fig. 2 demonstrated that the absorbance value of E11 decreased lightly as the concentration of the DP increased, while it kept relatively stable for 293 T cells, which might due to the positive charge of the AMP affected the integrity of part E11 cells moderately. The survival rates of E11 and 293 T cells were greater than 80% and 90% respectively. Cell morphology of E11 and 293 T were also investigated by a fluorescence inverted microscopy, images showed that the morphology of the cells has not changed significantly, proved that DP has not toxicity to zebrafish cells and human epithelial cells and is an environmentally friendly AMP (Fig S3 and Fig S4).

Peptide modified metal surface was generated by depositing DP onto 304 SS. The functional groups of SS-DP surface were collected by FTIR to prove the presence of DP on the SS surface. FTIR spectrum of SS-DP surface was shown in Fig. 3 with SS as background. The peak at 3300 ~ 3500 cm^{−1} was contributed to Amide A of peptide, which demonstrated the existence of −NH₂, −NH, and other related functional groups. Two peaks appeared at 2800 ~ 3000 cm^{−1} were derived from the C-H stretching vibration of ν (C-H) (including saturated or unsaturated). The two strong peaks of DP were observed at 1750 cm^{−1} (amide I, C = O stretching) and 1650 cm^{−1} (amide II, C-N stretching and N-H bending) [46]. O-C-O stretching of ester corresponded to the peak at around 1280 cm^{−1}, while the peak at 1460 cm^{−1} is the bending vibration of −CH [47]. The two peaks around 1100 cm^{−1} were C-O stretching of C-OH, and = C-H stretching peak can be observed at around 850 cm^{−1} [46]. The functional groups corresponding to the above peaks do not belong to 304

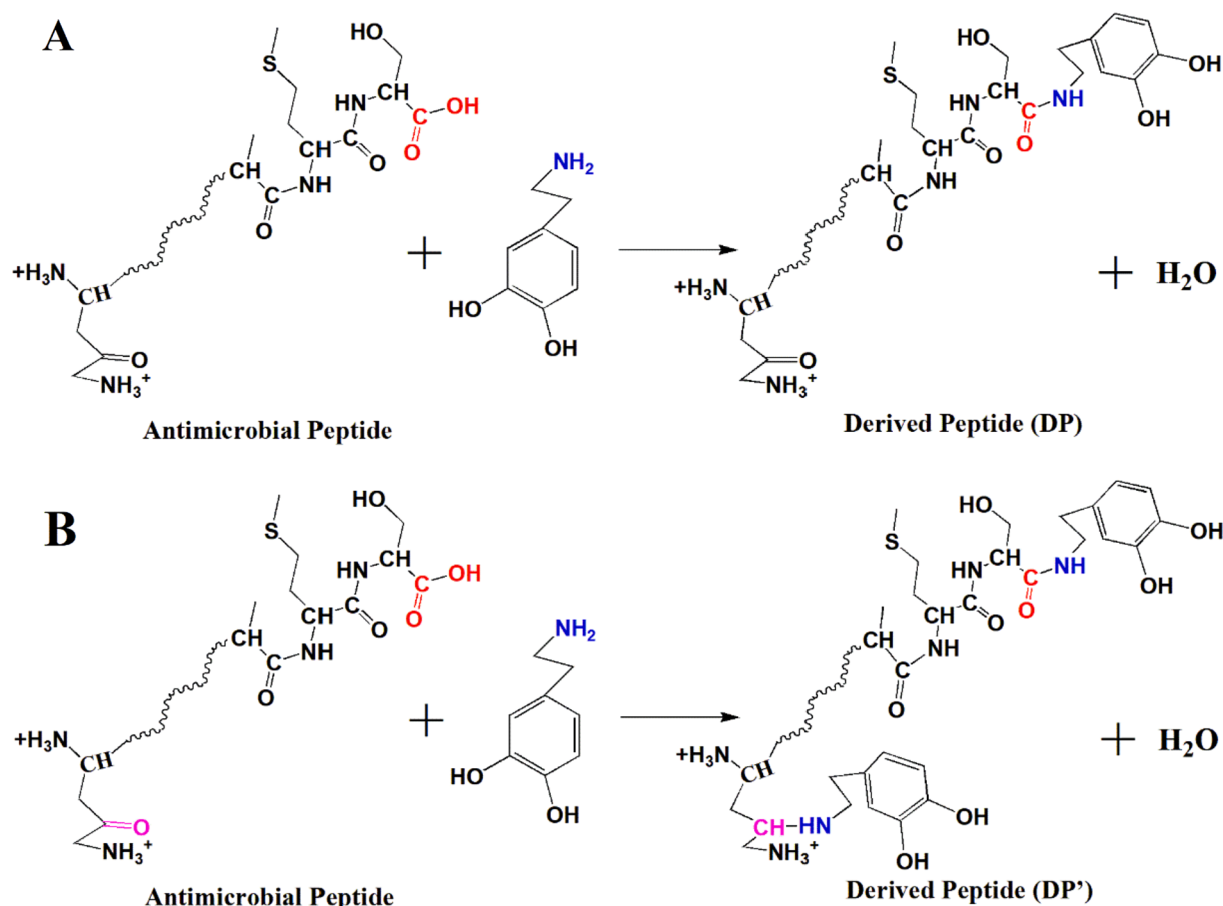


Fig. 1. Schematic illustration of the synthesis of DP (A) and DP' (B).

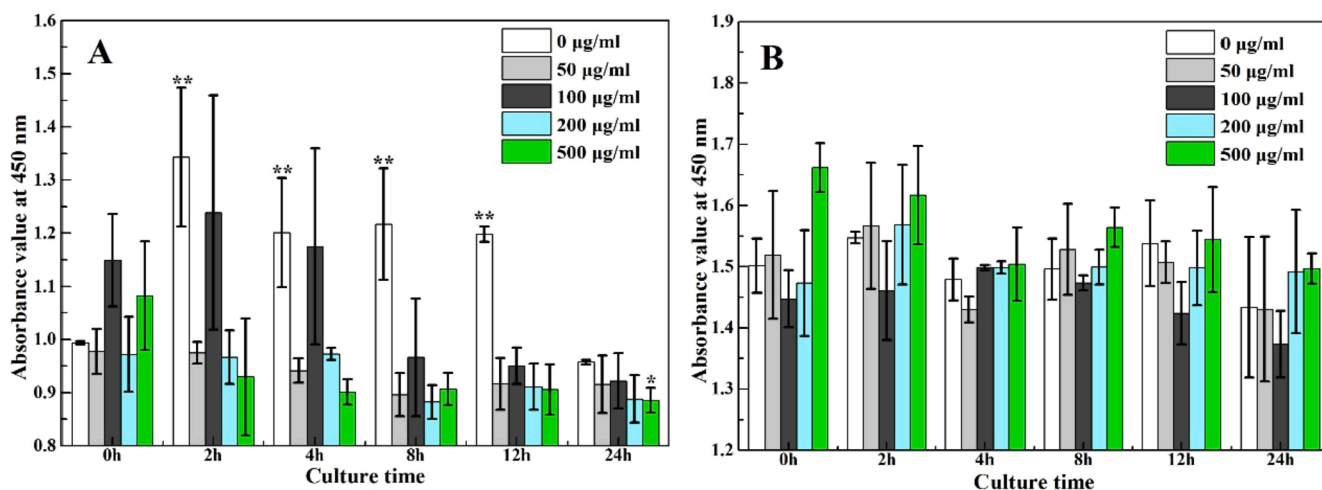


Fig. 2. Value-added of cells after incubation with different concentrations of DP for 0, 2, 4, 8, 12, and 24 h. A: E11; B: 293 T. (** $p < 0.01$, * $0.01 \leq p < 0.05$).

SS, which proved the existence of DP on the surface of 304 SS.

The element's chemical information of SS-DP surface was detected by XPS, and the XPS spectra shown the evidence that DP was bonded onto 304 SS surface. It should be noted that the untreated AMP does not contain disulfide bonds or indole groups, leading to unsuccessful directly reaction between AMP and SS surface. The oxygen content on the SS surface was extremely high, and the C/O content ratio was about 0.58. This result was due to the oxidation of the metal surface under atmospheric environment, which greatly increased oxygen content of the sample surface. After DP grafting, iron percentage of SS-DP surface

was lower than SS surface significantly, while the percentage of element sulfur increased due to modification of DP (Table S1). Oxygen and nitrogen contents of surface significantly increased because of the modification of DP, among which the increase ratio of nitrogen content was greater. The C/O ratio of sample surface increased from 0.58 to 2.51, while the C/N ratio dropped sharply from 40.45 to 4.02 (Fig. 4A). The above element amount changes were due to XPS test depth was about 2–10 nm, and the element ratio on the surface after DP modification was similar to that of organic matter. Fig. 4B demonstrated the XPS spectra C1s peaks of SS and SS-DP surface, three peaks at 288.1, 286.2, and

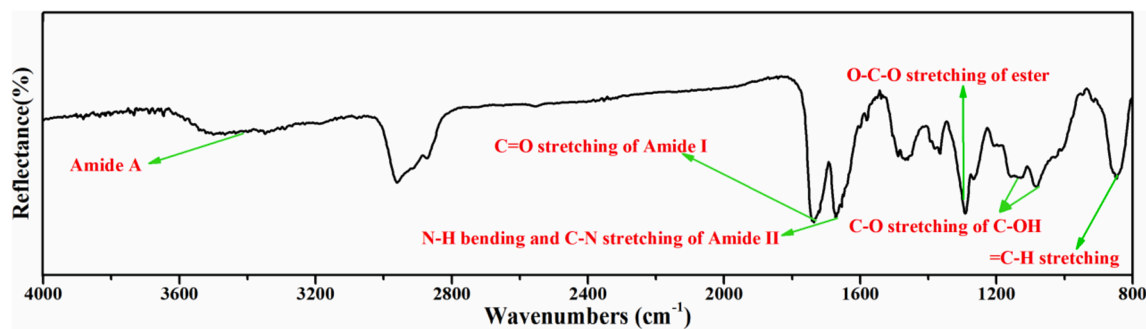


Fig. 3. FTIR of DP treated sample with SS as background.

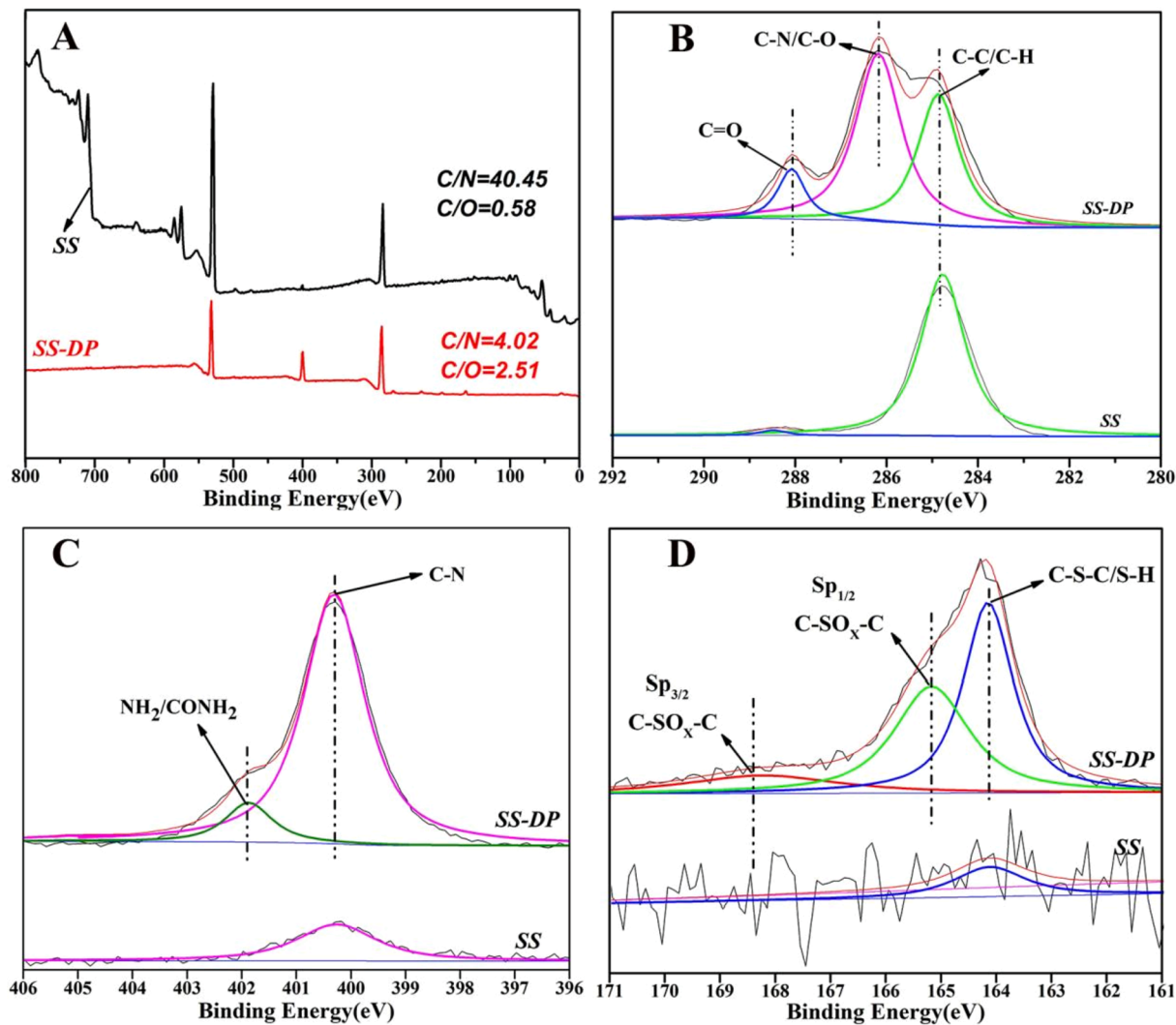


Fig. 4. X-ray photoelectron spectra of the SS surfaces and SS-DP surface. A: Full spectrum; B: C 1 s; C: N 1 s; D: S 2p.

284.6 eV contributed by N-C=O of derived peptide, carbon atoms in N-C/C-O bonds, and carbon C-C/C-H [48]. The N1s spectra of the samples were showed that more peaks appeared in SS-DA surface, the peak at 401.9 eV was caused by function group of O=C-NH₂ and -NH₃⁺/-NH₂, and the peak at around 400.8 eV was C-N (Fig. 4C) [49]. Fig. 4D illustrated the XPS spectra S 2p peaks of SS and SS-DP surfaces, two additional peaks appeared at 168.4 and 165.1 eV of SS-DP surface compared to SS surface, which attributed to Sp_{1/2}C-SO_x-C and Sp_{3/2}C-SO_x-C of DP respectively [50,51]. From the above analysis, it can be seen that DP was successfully grafted onto the SS surface after the DP modification, and

derived peptide treated surface was covered with a layer of DP, which made the contents of N, C and S of SS surface increased to different degrees. Because of the insufficient depth of XPS measurement, the amount of Fe was almost negligible, and the spectrum results of the elements proved that DP was successfully grafted onto the surface of stainless steel.

Contact angle was usually used to evaluate wettability of surface, and Fig. 5 showed the contact angle of SS samples before and after DP modification. The contact angle of sample surfaces decreased from 69.4° ± 0.8° to 40.7° ± 1.8° after DP treatment, which caused by the well

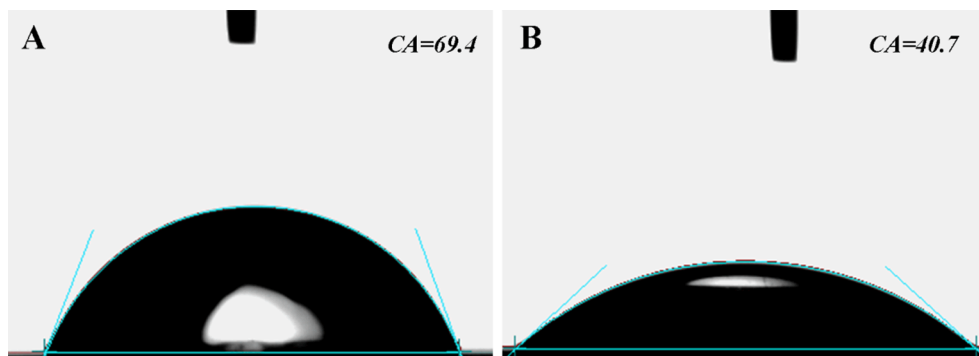


Fig. 5. Contact angles of substrate surfaces. A: SS surface; B: SS-DP surface.

hydrophilic properties of dopamine residues and oxidative polymerized productions of DP. The grand average of hydropathicity (GRAVY) of peptide represent its hydrophilic properties, decreasing negative score indicated greater hydrophilicity. The GRAVY value of untreated AMP is -1.171 , which illustrated it is hydrophilic peptide. Surface Contact angle was reduced significantly under the combined effect of hydrophilic AMP and dopamine. The change in contact angles on the surfaces of SS and SS-DA indicated that DP was successful grafted onto 304 SS surface.

The surface morphology of the samples was characterized via AFM and a 3D optical profilometer. Fig. 6 demonstrated the surface morphology of 304 SS and SS-DP. The SS surface smoother than SS-DP surface significantly, and maximum vertical height increased from 6.4 nm to 10.18 nm after DP modification. After the DP modification, the polymerization of dopamine residues on the DP increased the maximum vertical height of the sample surface. The AFM image can only show the surface topography of a limited area. 3D optical profilometer was used to collect the topography of the sample surface for obtaining the surface macro topography of SS and SS-DP. Fig. 7 demonstrated surface morphology of SS and SS-DP, 3D profile of surfaces in X direction and Y direction decreased from 0.4202 μm and 0.2254 μm to 0.1477 μm and 0.1805 μm respectively after DP modification. The troughs of SS surface were filled into DP particles, reducing the roughness of the sample surface, and the surface roughness values (R_a) decreased from 237.81 ± 15.04 nm to 122.75 ± 10.97 nm (Table S2). R_z refers to the sum of the average value of the 5 maximum profile peak heights and the 5 maximum profile valley depths within the sampling length, and it reflects the height of the sample surface profile. R_z values changed from 3.903 ± 0.197 μm to 3.978 ± 0.246 μm (Table S2), and their R_z values did not change significantly. All changes in surface morphology confirmed successful grafting of the derived antimicrobial peptide. The

thickness of DP was detected by an ellipsometry, the thickness of the organic matter on the surface of substance has increased significantly. The thickness of organic compositions of SS-DP was 13.49 ± 1.80 nm compared to 1.70 ± 1.77 nm of SS surface (Fig S5), which caused by the grafting of derived peptide. Fig. 6S showed coating thickness change on the sample surface, the coating thickness on the sample surface decreased to 12.51 ± 1.58 nm due to unstable binding of macromolecular antimicrobial peptide derivative polymers after 7 days immersion, and it tends to stabilize to 12.28 ± 1.67 nm after 14 days immersion, which suggested that the antifouling coating bonded on the surface firmly.

The antifouling performance of derived AMP modified surface were assessed by bacterial and algal adhesion tests. *V. natriegens* and *P. tricornutum* were selected to conduct adhesion assays, which because they are typical marine bacteria and seaweeds in the East China Sea respectively. The adhesion of bacteria were evaluated by CLSM and SEM, Fig. 8 showed the adhesion of *V. natriegens* on the surface of SS and SS-DP. A large number of bacteria adhere to the surface of the stainless steel sample, the amount of attached bacteria decreased significantly after DP modification (Fig. 8A, Fig. 8B). The attached bacteria were stained into red by PI dye, the coverage percentages of *V. natriegens* were calculated to 6.112% and 0.136% on SS and SS-DP surfaces. Bacterial attachment ratio decreased 97.78% , which illustrated that derived peptide treated surface has strong antibacterial property because of antibacterial performance of AMPs. SEM images showed a similar result, and a few bacteria still attached on substance, while almost all attached bacteria were damaged (Fig. 8C, Fig. 8D). The damaged bacterial cell wall may be result from electrostatic reaction between DP residues and phospholipids in bacterial biofilms and entry of DP residues into the bacterial cell to kill the bacteria [52]. AMPs are usually cationic peptides, while the phospholipid layers of bacterial cells are negatively

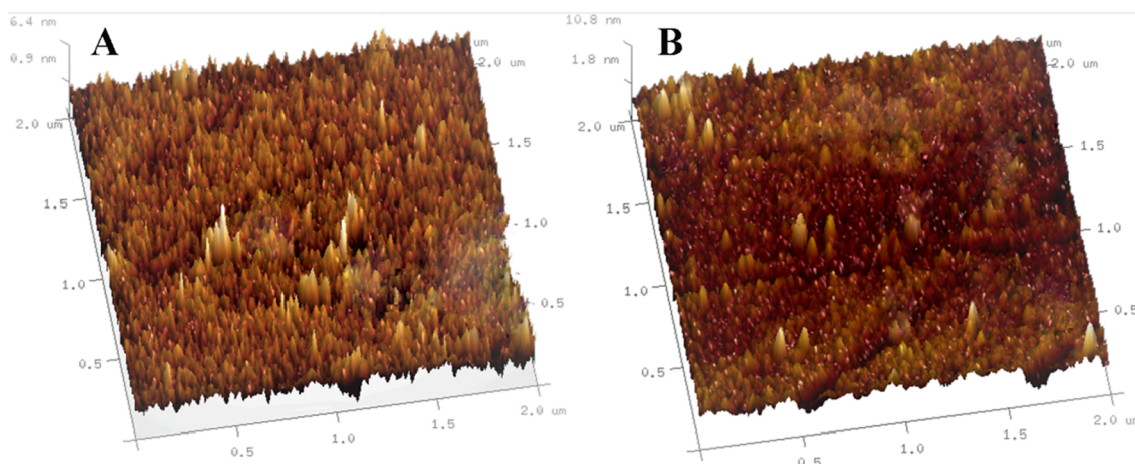


Fig. 6. AFM 3D images of the SS surface (A) and SS-DP surface (B).

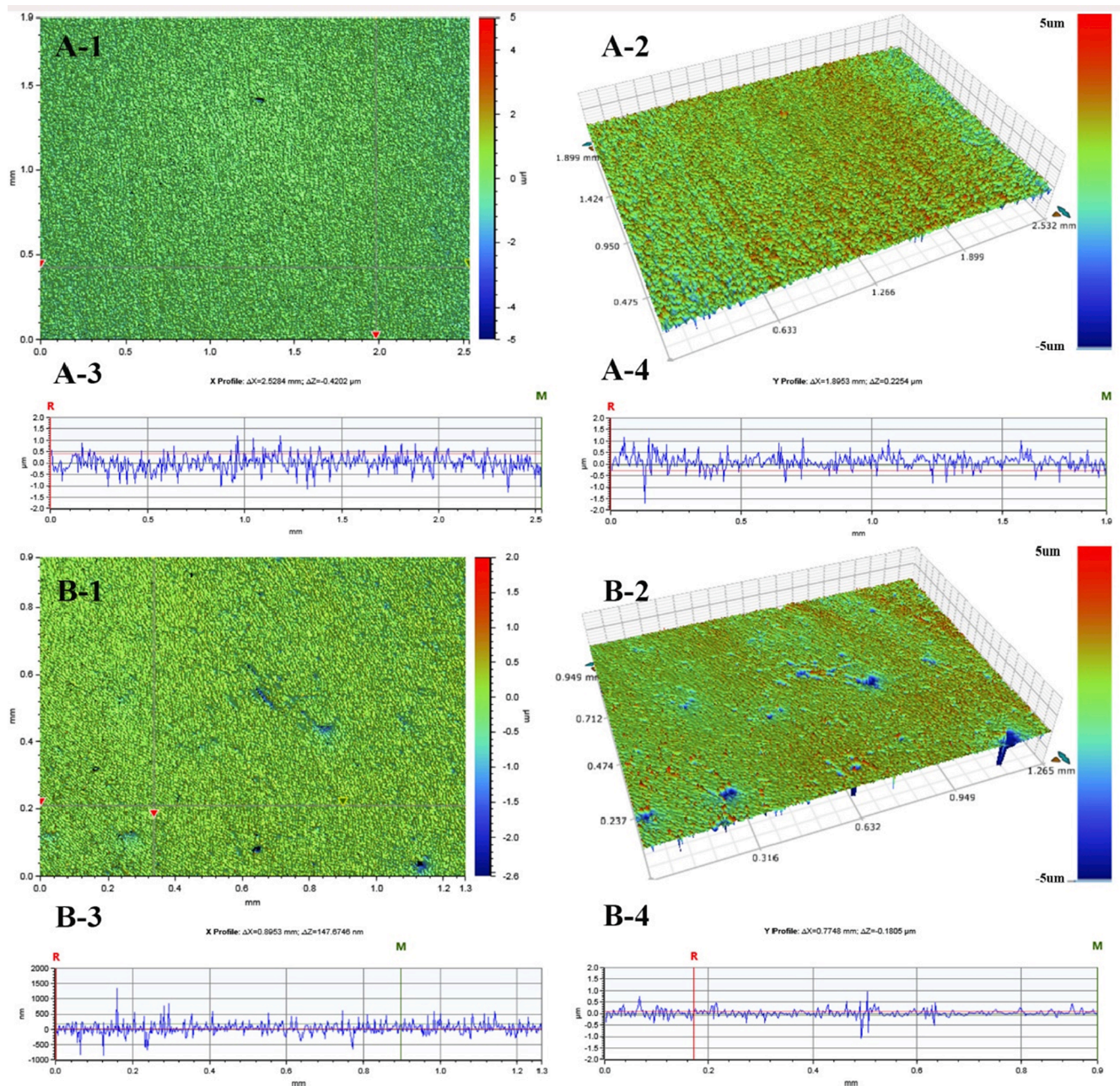


Fig. 7. Surface morphology of SS (A) and SS-DP (B). –1: main view of images; –2: 3D images; –3: X axis profile; –4: Y axis profile.

charged [53]. Bacteria were killed due to destroyed integrity of bacterial cell when the positive charged antimicrobial peptide derivatives contact the negative charged phospholipid layer of the bacterial outer membrane [54]. The –SH with negative charge of DP will also stop bacterial adhesion due to electrostatic repulsion [55].

Anti-algal test is another method to evaluate antifouling performance of surface. Less algae attached to the surface of SS-DP compared to SS after 7 days immersion in *P. tricornutum* solution (Fig. 9). CLSM images of SS and SS-DP illustrated that *P. tricornutum* coverage of substance decreased significantly. DP caused the algal adhesion ratio decrease from 30.31% to 3.02%, and the reduction ratio reached 90.01%. SS-DP surface will form a hydration layer after being immersed into algal solution. The hydrolyzed DP residues in the hydration layer entered the algae cells and triggered the leakage of intracellular substance from algal cells, then algae were killed [9,56].

The antifouling robustness of antimicrobial peptide coated surface was assessed by immersing SS-DP into sterilized ASW for 1, 7, 14 and 21

days with gentle shaking (30 rpm). The antifouling performance of SS-DP after immersion in ASW were tested by *V. natriegens* and *P. tricornutum* attachment assays. The antibacterial capacity of SS-DP decreased after 7 days immersion, while it tends to be stable after 14 days immersion. Compared to surface after 7 days immersion by ASW, the anti-algal performance of 14 days treated SS-DP surface decreased more than 5%, antibacterial efficiency tends to be stable after 14 days immersion (Fig. 10). Most of the DP and its polymer were bonded to the substrate to form a long-lasting antifouling surface during the immersion process, and part of the macromolecular antimicrobial peptide derivative polymer that was not firmly bound or unstable in seawater detached from the substrate, reducing the antifouling performance of the surface.

Tafel polarization curves have been widely used to evaluate the efficiency of metal surface coatings against biocorrosion [57]. The determined inhibition efficiency (IE) values were obtained by analyzing Tafel plots of the original SS and DP modified SS samples immersed in

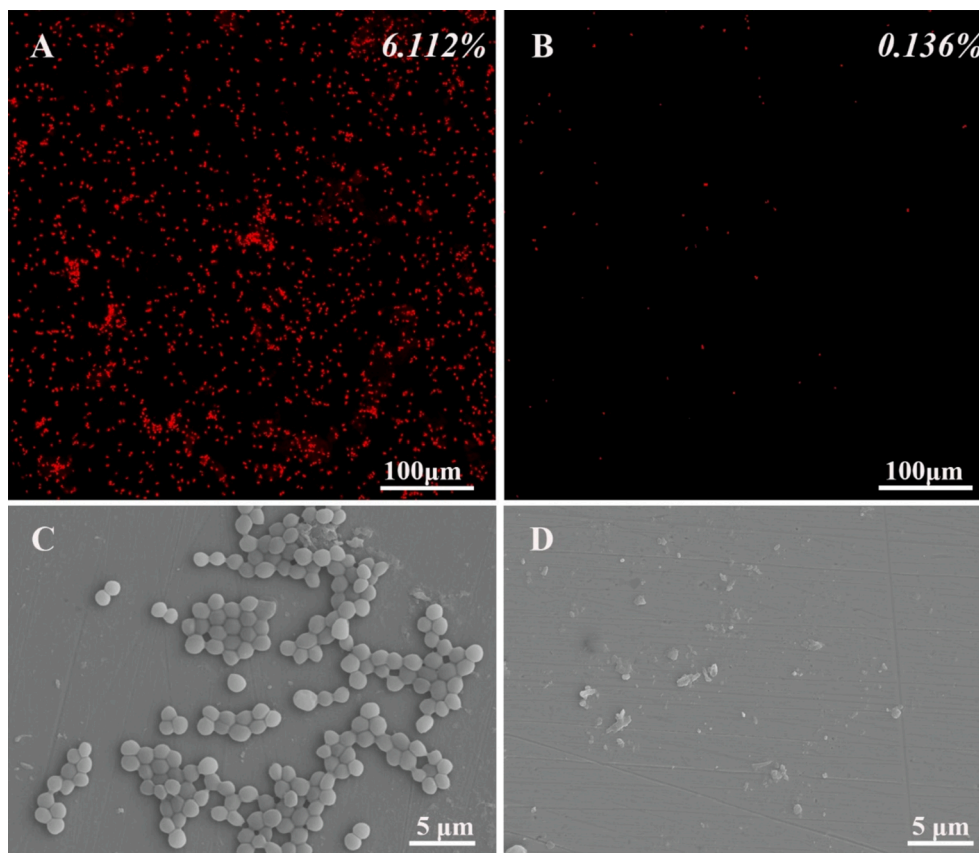


Fig. 8. CLSM images of *V. natriegens* attached to surfaces of SS (A) and SS-DP (B), SEM images of *V. natriegens* attached to surfaces of SS (C) and SS-DP (D).

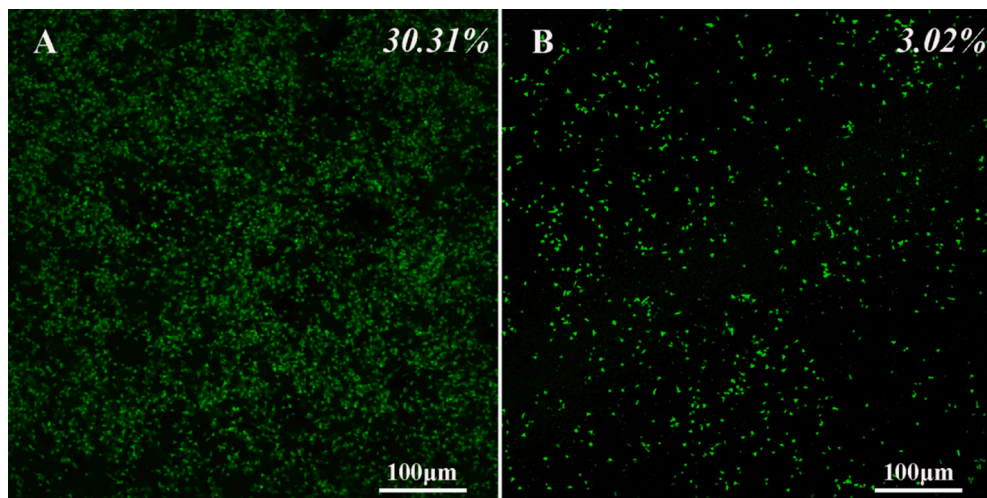


Fig. 9. CLSM images of *P. tricornutum* attached to surfaces of SS (A) and SS-DP (B).

V. natriegens solutions for 1, 3, and 7 days, which were used to evaluate the corrosion resistance of the samples before and after modification. Fig. 11 showed the Tafel plots of the original and DP modified samples after 1, 3, and 7 days of exposure to *V. natriegens* solutions. Compared with 1 day, the polarization curves of both SS and SS-DP samples shifted downward at 3 and 7 days, and the corrosion potential (E_{corr}) decreased. The polarization curves of SS samples had a significant shift to the right at 3 and 7 days compared with 1 day, while the polarization curves of SS-DP samples at 3 days and 7 days had a slight shift to the right, indicating that the corrosion current density increased, but the corrosion current density of the SS-DP increased less rapidly than that of the original

samples. With the increasing of soaking time, *V. natriegens* attached to the surface of the samples to multiply, breaking the integrity of the passivation film on the surface of the samples and reducing the corrosion resistance of the samples [58]. However, the grafting of antimicrobial peptides in the SS surface inhibited the adhesion of *V. natriegens*, making the corrosion current significantly smaller than that of the unmodified samples.

The parameters of the polarization curves of the samples before and after the modification were obtained by fitting the Tafel extrapolation method by pushing the anodic and cathodic branches to the intersection, as shown in Table 1. The E_{corr} values of the SS samples showed a

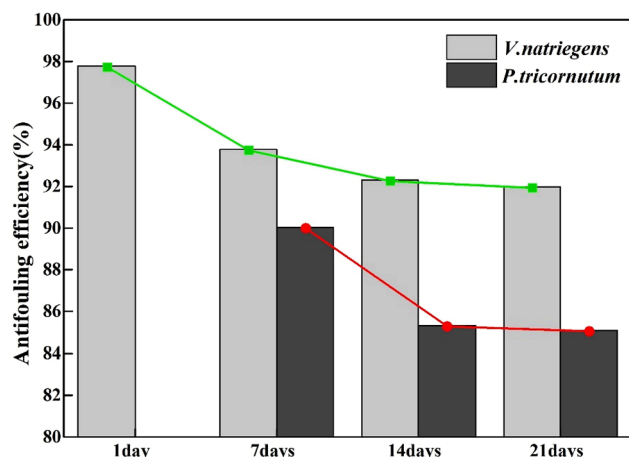


Fig. 10. Robust of antifouling efficiency of SS-DP surfaces.

decreasing trend from -0.402 V at 1 day to -0.411 V and -0.442 V at 3 days and 7 days respectively. The i_{corr} values showed an increasing trend from $0.887 \mu\text{A}\cdot\text{cm}^{-2}$ in the 1 day to $1.190 \mu\text{A}\cdot\text{cm}^{-2}$ in the 3 days, and increasing to $1.350 \mu\text{A}\cdot\text{cm}^{-2}$ in the 7 days, which means that the corrosion rate continued to increase. In contrast, the corrosion potential of SS-DP samples shifted negatively from -0.300 V in the 1 day to -0.306 V in the 3 days and then to -0.345 V in the 7 days. The corrosion current density showed a tendency to first increase and then remain relatively constant, increasing from $0.159 \mu\text{A}\cdot\text{cm}^{-2}$ in the 1 day to $0.306 \mu\text{A}\cdot\text{cm}^{-2}$ in the 3 days, in 7 days increasing to $0.392 \mu\text{A}\cdot\text{cm}^{-2}$, the corrosion rate was significantly less than that of the original sample. The fitted data were consistent with the polarization graph, with the increase of immersion time, the sample surface was corroded to break the ring, the corrosion potential was negatively shifted, the corrosion current density increased, and the corrosion of the sample surface was accelerated. However, the corrosion potential of the samples grafted with antimicrobial peptides was always larger than that of the SS samples, and the corrosion rate was also significantly less than the SS samples, which may be due to the successful grafting of antimicrobial peptides increased the electron escape work on the sample surface and reduced the electron activity on the sample surface, reducing the corrosion energy on the material surface finally [59]. The i_{corr} values of SS-DP samples were reduced 4-fold and 3-fold relative to SS samples exposed *V. natriegens* solutions for 3 and 7 days, respectively, after exposure in medium inoculated with *V. natriegens*. Moreover, the inhibition efficiency (IE) of SS-DP samples compared to SS samples exposed for 3 and 7 days reached more than 74% and 70%, respectively. The above analysis indicated that the modified samples have excellent corrosion resistance.

4. Conclusions

In this study, a derived AMP with strong biocompatibility was synthesized by conjugation of AMP with dopamine in weak acid solution, and then the derived AMP was utilized to modify 304 SS surface to generate an antifouling surface. FTIR and XPS results showed that DP was bound onto 304 SS surface successfully, and their wettability and morphology were also changed after modification. Adhesion tests of *V. natriegens* and *P. tricornutum* on 304 SS and DP treated surfaces demonstrated that SS-DP surface possessed strong and long-lasting antifouling performance. The bactericidal and anti-algae properties of the SS-DP surface can be considered as the synergistic result of destroying the integrity of bioorganisms cells by electrostatic interaction between DP and bioorganisms cells and the interaction of DP residues into the bioorganisms cells. Electrochemical experiments showed that the DP modified surface had a slower corrosion rate compared to the initial sample, and still showed an inhibition efficiency of over 70% during a week of incubation in *V. natriegens* solutions, demonstrating the excellent corrosion resistance of the modified sample surface. The novel approach can be used to prepare such AMP-tethered antifouling and biocompatible SS substrates having potential usage in marine equipment, biomedical devices and other applications.

CRedit authorship contribution statement

Pan Cao: Conceptualization, Methodology, Data curation, Writing – original draft, Funding acquisition. **De Liu:** Methodology, Investigation, Data curation. **Yanbing Zhang:** Investigation, Data curation. **Fei Xiao:** Investigation, Data curation. **Chengqing Yuan:** Writing – review & editing. **Feng Liang:** Investigation, Data curation. **Xiaodan Liu:** Methodology, Writing – review & editing. **Chao Zhang:** Supervision, Funding acquisition.

Table 1

Analyzed parameters of Tafel plots of SS and SS-DP after exposure to *Vibrio natriegens* inoculation medium for 1, 3 and 7 days.

Time/ d	samples	E_{corr}/V	$i_{\text{corr}}/\mu\text{A}\cdot\text{cm}^{-2}$	B_a/mV	B_c/mV	CR/ $\text{mm}\cdot\text{y}^{-1}$	IE/%
1	SS	-0.402	0.887	563	118	2.016	
	SS-DP	-0.300	0.159	55	192	0.361	82.07
3	SS	-0.411	1.190	687	111	2.710	
	SS-DP	-0.306	0.306	87	131	0.763	74.28
7	SS	-0.442	1.350	724	69	3.081	
	SS-DP	-0.345	0.392	53	118	0.892	70.96

E_{corr} denotes the corrosion potential. B_a denotes the anodic Tafel slope. B_c denotes the cathodic Tafel slope. CR denotes corrosion rate. IE denotes inhibition efficiency.

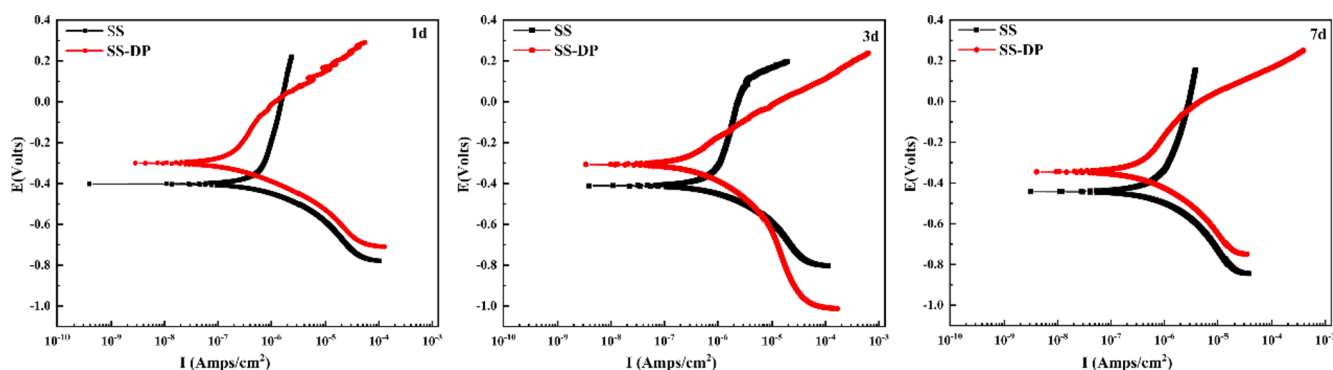


Fig. 11. Tafel plots of the SS, SS-DP in the *Vibrio natriegens* inoculated medium for 1 day (A), 3 days (B), and 7 days (C).

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors gratefully acknowledge the Natural Science Foundation of Jiangsu Province (No. BK20190916), the financial support of the National Natural Science Foundation of China (No. 51905468), 'Blue Project' of Yangzhou University and the Yangzhou City-Yangzhou University Cooperation Foundation (No. YZU201801).

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.apsusc.2022.153019>.

References

- [1] S.u. Xin, M. Yang, D. Hao, X. Guo, L. Jiang, Marine antifouling coatings with surface topographies triggered by phase segregation, *J. Colloid Interface Sci.* 598 (2021) 104–112.
- [2] S. Majhi, V.C. Peddiraju, A. Mishra, Effect of antimicrobial peptide (AMP)-tethered stainless steel surfaces on the bacterial membrane, *Mater. Today Chem.* 21 (2021), 100541.
- [3] I. Amara, W. Mileda, R.B. Slamab, N. Ladhari, Antifouling processes and toxicity effects of antifouling paints on marine environment A review, *Environmental Toxicology Pharmacology* 57 (2018) 115–130.
- [4] X. He, XiuqinBai FengTian, ChunWang ChengqingYuan, A. Neville, Biomimetic lubricant-infused titania nanoparticle surfaces via layer-by-layer deposition to control biofouling, *Appl. Surf. Sci.* 515 (2020), 146064.
- [5] Avishek Kumar, Ahmed AL-Jumaili, Olha Bazaka, Elena P. Ivanova, Igor Levchenko, Kateryna Bazaka, Mohan V. Jacob. Functional nanomaterials, synergisms, and biomimicry for environmentally benign marine antifouling technology. *Materials Horizons*, 2021, 8: 3201–3238.
- [6] A.M. Maan, A.H. Hofman, W.M. de Vos, M. Kamperman, Recent developments and practical feasibility of polymer-based antifouling coatings, *Adv. Funct. Mater.* 30 (2020) 2000936.
- [7] N. García-Bueno, A. Marín, Ecological management of biomass and metal bioaccumulation in fish-cage nettings: Influence of antifouling paint and fiber manufacture, *Aquaculture* 544 (2021), 737142.
- [8] Y. Xin, Y. Yang, W. Yang, X. Wang, X. Liu, F. Zhou, Y. Zhao, One-step zwitterionization and quaternization of thick PDMAEMA layer grafted through subsurface-initiated ATRP for robust antibiofouling and antibacterial coating on PDMS, *J. Colloid Interface Sci.* 610 (2022) 234–245.
- [9] Y. Zou, L. Kunyan, Y. Lin, W. Yan, Y. Wang, L. Li, C. Huang, Y. Zhang, J.L. Brash, H. Chen, Y. Qian, Dual-functional surfaces based on an antifouling polymer and a natural antibiofilm molecule: Prevention of biofilm formation without using biocides, *ACS Appl. Mater. Interfaces* 13 (2021) 45191–45200.
- [10] Y. Kai, A. Alzahrani, S. Khoddami, J.T. Cheng, Y. Mei, A. Gill, H.D. Luo, E.F. Haney, K. Hilpert, R.E. Hancock, D. Lange, J.N. Kizhakkedathu, Rapid assembly of infection-resistant coatings: Screening and identification of antimicrobial peptides works in cooperation with an antifouling background, *ACS Appl. Mater. Interfaces* 13 (2021) 36784–36799.
- [11] V. Ortiz-Gómez, V.D. Rodríguez-Ramos, R. Maldonado-Hernández, J.A. González-Feliciano, E. Nicolau, Antimicrobial polymer-peptide conjugates based on maximin H5 and PEG to prevent biofouling of *E. coli* and *P. aeruginosa*, *ACS Appl. Mater. Interfaces* 12 (2020) 46991–47001.
- [12] P. Cao, D. Chongwei, X. He, C. Zhang, C. Yuan, Modification of a derived antimicrobial peptide on steel surface for marine bacterial resistance, *Appl. Surf. Sci.* 510 (2020), 145512.
- [13] Seong-Cheol Park, Jong-Kook Lee, Si Wouk Kim, Yoonkyung Park. Selective algicidal action of peptides against harmful algal bloom species. *PLoS One*, 2011, 6: e26733.
- [14] W.Y. Wong, A. Patricia Campbell, C. McInnes, B.D. Sykes, W. Paranchych, R. T. Irvin, R.S. Hodges, Structure-function analysis of the adherence-binding domain on the pilin of *Pseudomonas aeruginosa* strains PAK and KB7, *Biochemistry* 34 (1995) 12963–12972.
- [15] E.M. Davis, D.-Y. Li, R.T. Irvin, A peptide-stainless steel reaction that yields a new bioorganic-metal state of matter, *Biomaterials* 32 (2011) 5311–5319.
- [16] P. Cao, C. Yuan, J. Xiao, X. He, X. Bai, A biofilm resistance surface yielded by grafting of antimicrobial peptides on stainless steel surface, *Surf. Interface Anal.* 50 (2018) 516–521.
- [17] A. Mondal, P. Singha, M. Douglass, L. Estes, M. Garren, L. Griffin, A. Kumar, H. Handa, A synergistic new approach toward enhanced antibacterial efficacy via antimicrobial peptide immobilization on a nitric oxide-releasing surface, *ACS Appl. Mater. Interfaces* 13 (2021) 43892–43903.
- [18] M. Xiao, J. Jasensky, L. Foster, K. Kuroda, Z. Chen, Monitoring antimicrobial mechanisms of surface-immobilized peptides in situ, *Langmuir* 34 (2018) 2057–2062.
- [19] M. Xiao, J. Jasensky, J. Gerszberg, J. Chen, J. Tian, T. Lin, L.u. Tieyi, J. Lahann, Z. Chen, Chemically immobilized antimicrobial peptide on polymer and self-assembled monolayer substrates, *Langmuir* 34 (2018) 12889–12896.
- [20] Vincent Humblot 1, Jean-Fabrice Yala, Pascal Thebault, Kada Boukerma, Arnaud Héquet, Jean-Marc Berjeaud, Claire-Marie Pradier. The antibacterial activity of Magainin I immobilized onto mixed thiols self-assembled monolayers. *Biomaterials*, 2009, 30: 3503–3512.
- [21] D. Dutta, N. Kumar, M.D. Willcox, Antimicrobial activity of four cationic peptides immobilised to poly-hydroxyethylmethacrylate, *Biofouling* 32 (2016) 429–438.
- [22] E. Faure, P. Lecomte, S. Lenoir, C. Vreuls, C. Van De Weerd, C. Archambeau, J. Martial, C. Jérôme, A.-S. Duwez, C. Detrembleur, Sustainable and bio-inspired chemistry for robust antibacterial activity of stainless steel, *J. Mater. Chem.* 21 (2011) 7901–7904.
- [23] C. Vreuls, G. Zocchi, B. Thierry, G. Garitte, S.S. Griesser, C. Archambeau, C. Van de Weerd, J. Martial, H. Griesser, Prevention of bacterial biofilms by covalent immobilization of peptides onto plasma polymer functionalized substrates, *J. Mater. Chem.* 20 (2010) 8092–8098.
- [24] Y. Liu, K. Ai, L.u. Lehui, Polydopamine and its derivative materials: Synthesis and promising applications in energy, environmental, and biomedical fields, *Chem. Rev.* 114 (2014) 5057–5115.
- [25] N. Nazi, V. Humblot, C. Debiemme-Chouvy, A new antibacterial N-Halamine coating based on polydopamine, *Langmuir* 36 (2020) 11005–11014.
- [26] I.I. Niyonshuti, V.R. Krishnamurthi, D. Okyere, L. Song, M. Benamara, X. Tong, Y. Wang, J. Chen, Polydopamine surface coating synergizes the antimicrobial activity of silver nanoparticles, *ACS Appl. Mater. Interfaces* 12 (2020) 40067–40077.
- [27] Jeongwoo Hong, Sangdon Choi, Dong G. Jwa, Min Kim, Sung Min Kang. Mussel-inspired, one-step thiol functionalization of solid surfaces. *Langmuir*, 2020, 36: 1608–1614.
- [28] J. Wang, J. Tian, S. Gao, W. Shi, F. Cui, Dopamine triggered one step polymerization and codeposition of reactive surfactant on PES membrane surface for antifouling modification, *Sep. Purif. Technol.* 249 (2020), 117148.
- [29] I.B. Pinto, L.S. Machado, B.T. Meneguetti, M.L. Nogueira, C.M. Carvalho, A.R. Roel, O.L. Franco, Utilization of antimicrobial peptides, analogues and mimics in creating antimicrobial surfaces and bio-materials, *Biochem. Eng. J.* 150 (2019), 107237.
- [30] P. Cao, X. He, J. Xiao, C. Yuan, X. Bai, Covalent bonding of AgNPs to 304 stainless steel by reduction in situ for antifouling applications, *Appl. Surf. Sci.* 452 (2018) 201–209.
- [31] W.-B. Tsai, C.-Y. Chien, H. Thissen, J.-Y. Lai, Dopamine-assisted immobilization of poly(ethylene imine) based polymers for control of cell-surface interactions, *Acta Biomater.* 7 (2011) 2518–2525.
- [32] Q. Zeng, Y. Zhu, Y.u. Bingran, Y. Sun, X. Ding, X.u. Chen, W.u. Yu-Wei, Z. Tang, X. u. Fu-Jian, Antimicrobial and antifouling polymeric agents for surface functionalization of medical implants, *Biomacromolecules* 19 (2018) 2805–2811.
- [33] R. Wanka, N. Aldred, J.A. Finlay, A. Amuthalingam, J.L. Clarke, A.S. Clare, A. Rosenhahn, Antifouling properties of dendritic polyglycerols against marine macrofouling organisms, *Langmuir* 35 (2019) 16568–16575.
- [34] G. Wei, D. Jiang, H.u. Shuai, Z. Yang, Z. Zhang, W. Li, W. Cai, D. Liu, Polydopamine-decorated microcomposites promote functional recovery of an injured spinal cord by inhibiting neuroinflammation, *ACS Appl. Mater. Interfaces* 13 (2021) 47341–47353.
- [35] C.-T. Chen, F.J. Martin-Martinez, G.S. Jung, M.J. Buehler, Polydopamine and eumelanin molecular structures investigated with *ab initio* calculations, *Chem. Sci.* 8 (2017) 1631–1641.
- [36] V. Maselli, E. Galdiero, A.M. Salzano, A. Scaloni, A. Maione, A. Falanga, D. Naviglio, M. Guida, A.D. Cosmo, S. Galdiero, Octopartepin: Identification and preliminary characterization of a novel antimicrobial peptide from the suckers of *Octopus vulgaris*, *Mar. Drugs* 18 (2020) 380.
- [37] M. Tonelli, I. Perini, F. Ridi, B. Piero, Improving the properties of antifouling hybrid composites: The use of Halloysites as nano-containers in epoxy coatings, *Colloids Surf., A* 623 (2021), 126779.
- [38] F. Ye, W. Wang, L. Zhang, V. Vinokurov, A. Stavitskaya, Y. Lvov, Development of marine antifouling epoxy coating enhanced with clay nanotubes, *Materials* 12 (2019) 4195.
- [39] B. Zhang, Q. Yan, S. Yuan, X. Zhuang, F. Zhang, Enhanced antifouling and anticorrosion properties of stainless steel by biomimetic anchoring PEGDMA-Cross-Linking polycationic brushes, *Ind. Eng. Chem. Res.* 58 (17) (2019) 7107–7119.
- [40] H. Lee, S.M. Dellatore, W.M. Miller, P.B. Messersmith, Mussel-inspired surface chemistry for multifunctional coatings, *Science* 318 (2007) 426–430.
- [41] H. Wei, J. Ren, B. Han, X. Li, L. Han, L. Jia, Stability of polydopamine and poly (DOPA) melanin-like films on the surface of polymer membranes under strongly acidic and alkaline conditions, *Colloids and Surface B: Biointerfaces* 110 (2013) 22–28.
- [42] C. Zhang, M. Ma, T. Chen, H. Zhang, H. Dengfeng, W. Baiheng, J. Ji, X. Zhikang, Dopamine-triggered one-step polymerization and codeposition of acrylate monomers for functional coatings, *ACS Appl. Mater. Interfaces* 9 (2017) 34356–34366.
- [43] X. Zhang, Q. Huang, F. Deng, H. Huang, Q. Wan, M. Liu, Y. Wei, Mussel-inspired fabrication of functional materials and their environmental applications: Progress and prospects, *Appl. Mater. Today* 7 (2017) 222–238.

- [44] Y. Jeong, L.T. Thuy, S.H. Ki, S. Ko, S. Kim, W.K. Cho, J.S. Choi, S.M. Kang, Multipurpose antifouling coating of solid surfaces with the marine-derived polymer fucoidan, *Macromol. Biosci.* 18 (2018) 1800137.
- [45] P. Cao, X. He, J. Xiao, C. Yuan, X. Bai, Peptide-modified stainless steel with resistance capacity of *Staphylococcus aureus* biofilm formation, *Surf. Interface Anal.* 12–13 (2018) 1362–1369.
- [46] A.C. Talari, M.A. Martinez, Z. Movasaghi, S. Rehman, I.U. Rehman, Advances in fourier transform infrared (FTIR) spectroscopy of biological tissues, *Appl. Spectrosc. Rev.* 52 (2016) 456–506.
- [47] M. Carbonaro, A. Nucara, Secondary structure of food proteins by Fourier transform spectroscopy in the mid-infrared region, *Amino Acids* 38 (2010) 679–690.
- [48] C.-M. Pradier, M. Minier, M. Salmann, C. Méthivier, Covalent immobilization of lysozyme on stainless steel. Interface spectroscopic characterization and measurement of enzymatic activity, *Langmuir* 21 (2005) 5957–5965.
- [49] X. Kong, T. Kawai, J. Abe, T. Iyoda, Amphiphilic polymer brushes grown from the silicon surface by atom transfer radical polymerization, *Macromolecules* 34 (2001) 1837–1844.
- [50] T. Lou, X. Bai, X. He, C. Yuan, Antifouling performance analysis of peptide-modified glass microstructural surfaces, *Appl. Surf. Sci.* 541 (2021), 148384.
- [51] M. Wirde, U. Gelius, L. Nyholm, Self-assembled monolayers of cystamine and cysteamine on gold studied by XPS and voltammetry, *Langmuir* 15 (1999) 6370–6378.
- [52] W. Yandi, S. Mieszkina, A. di Fino, P. Martin-Tanchereau, M.E. Callow, J.A. Callow, L. Tyson, A.S. Clare, T. Ederth, Charged hydrophilic polymer brushes and their relevance for understanding marine biofouling, *Biofouling* 32 (2016) 609–625.
- [53] J. Yoo, A. Birke, J. Kim, Y. Jang, S.Y. Song, S. Ryu, B.-S. Kim, B.-G. Kim, M. Barz, K. Char, Cooperative catechol-functionalized polypeptide brushes and Ag nanoparticles for combination of protein resistance and antimicrobial activity on metal oxide surfaces, *Biomacromolecules* 19 (2018) 1602–1613.
- [54] W. Yang, P. Lin, D. Cheng, L. Zhang, W.u. Yang, Y. Liu, X. Pei, F. Zhou, Contribution of charges in polyvinyl alcohol networks to marine antifouling, *ACS Appl. Mater. Interfaces* 9 (2017) 18295–18304.
- [55] L. Zhong, Y. Song, S. Zhou, Covalent grafting of sodium p-styrene sulfonate to stainless steel for antibacterial applications, *Mater. Chem. Phys.* 268 (2021), 124753.
- [56] J. Zhang, S. Qian, L. Chen, L. Chen, L. Zhao, J. Feng, Highly antifouling double network hydrogel based on poly(sulfobetaine methacrylate) and sodium alginate with great toughness, *J. Mater. Sci. Technol.* 85 (2021) 235–244.
- [57] F. Mansfeld, B. Little, A technical review of electrochemical techniques applied to microbiologically influenced corrosion, *Corros. Sci.* 32 (3) (1991) 247–272.
- [58] D.E. Nivens, P.D. Nichols, J.M. Henson, G.G. Geesey, D.C. White, Reversible acceleration of the corrosion of AISI 304 stainless steel exposed to seawater induced by growth and secretions of the marine bacterium *Vibrio Natriegens*, *Corrosion* 42 (1986) 204–210.
- [59] B. Bharat, G.V. Anton, Measurements and analysis of surface potential change during wear of single-crystal silicon (100) at ultralow loads using Kelvin probe microscopy, *Appl. Surf. Sci.* 157 (4) (2000) 373–381.